Document 05

Calculation Canon

1. Purpose of This Document

This document defines the deterministic calculation rules for Trinity Engine v1.

Its purpose is to specify:

- the calculation sequence
- the order of operations
- the authoritative calculation logic for Astrology
- the authoritative calculation logic for Human Design
- the authoritative derivation logic for Gene Keys
- the authoritative derivation logic for Human Design structural outputs
- the treatment of node variants by domain
- the separation between internal calculation values and returned output objects

This document defines how Trinity Engine v1 must calculate supported outputs from canon compliant normalized input.

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2. General Calculation Principles

2.1 Deterministic Execution Rule

Given the same:

- normalized input payload
- canon version
- source stack version
- mapping version
- input schema version

the engine must always produce the same output payload or the same error payload.

No calculation step may depend on:

- hidden state
- runtime heuristics
- undocumented defaults
- external non pinned services
- manual judgment

- third party platform matching behavior

2.2 Order of Operations Rule

The engine must execute calculations in a fixed deterministic order.

No implementer may reorder steps if doing so could change output values, intermediate values, mapping boundaries, or derived structures.

If a later derivation depends on an earlier result, the earlier result must be treated as authoritative input for the later step.

2.3 Domain Separation Rule

The engine supports exactly three domains:

- Astrology
- Human Design
- Gene Keys

These domains share some underlying astronomical inputs, but they do not share output policy automatically.

This means:

- the same astronomical value may be used differently by different domains
- node policy is domain specific
- Human Design and Gene Keys derive from their own domain rules, not from Astrology output formatting
- Gene Keys is derived from Human Design output, not directly from an independent astronomical layer

2.4 Internal Value Rule

The engine may calculate internal values that are necessary to support canon defined outputs, even if those internal values are not returned as final output objects.

Internal values must still follow canon rules and must not leak into domain outputs unless explicitly allowed by the Output Specification.

Examples:

- True Node may be calculated internally while Mean Node remains authoritative for Astrology output
- UTC datetime may be calculated internally while local birth datetime remains part of input context
- Julian Day may be calculated internally without being returned in final output

2.5 No Silent Approximation Rule

The engine must not replace canon defined calculation methods with shortcuts, approximations, or convenience rules unless those approximations are explicitly pinned in the canon.

This includes but is not limited to:

- fixed day shortcuts instead of solar arc design time calculation
- alternate nodal variants
- alternate house systems
- alternate zodiac basis
- unpinned rounding shortcuts
- inferred structural derivations not explicitly defined in this document

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3. Shared Pre Calculation Steps

3.1 Input Validation

Before any calculation begins, the engine must validate the input payload according to Document 04.

If validation fails, no calculation may proceed.

3.2 Local Birth Datetime Construction

The engine must construct a local birth datetime from:

- `birth_date`
- `birth_time`
- `timezone`

This local datetime is the authoritative birth moment context at the input boundary.

3.3 UTC Conversion

The engine must convert the local birth datetime into UTC using the pinned timezone standard and canon compliant DST aware time handling.

This UTC birth moment is the authoritative astronomical reference moment for birth based calculations.

3.4 Julian Day Derivation

The engine must derive Julian Day or the equivalent canonical astronomical time input required by the pinned ephemeris source from the UTC birth moment.

This derived astronomical time reference is authoritative for the downstream ephemeris queries.

3.5 Birth Moment Coordinate Context

The engine must use the provided normalized:

- latitude
- longitude

as the authoritative birth location coordinates for all location dependent calculations, including house related calculations where applicable.

No place lookup, reverse lookup, or coordinate substitution is allowed.

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4. Core Mathematical and Angular Mechanics

4.1 Canonical Longitude Range

All ecliptic longitudes used by Trinity Engine v1 must be normalized to the canonical range:

- greater than or equal to `0.0`
- less than `360.0`

In interval notation, the canonical longitude domain is:

- `[0.0, 360.0)`

No supported longitude value may remain negative or equal to `360.0` after normalization.

4.2 Longitude Normalization Rule

If a raw longitude result is outside the canonical range, it must be normalized into `[0.0, 360.0)` by modular arithmetic.

The normalization rule is:

- add or subtract full 360 degree rotations until the result is within the canonical range
- `360.0` must normalize to `0.0`
- negative values must wrap forward into the canonical range

Examples:

- `-1.0` becomes `359.0`
- `360.0` becomes `0.0`
- `721.25` becomes `1.25`

This rule applies to:

- astronomical longitudes
- derived Earth longitudes
- derived South Node longitudes
- target solar arc longitudes for design time search
- any internal angular comparison used by the canon

4.3 Angular Opposition Rule

Where Earth or South Node is derived as the opposite of another object, the derivation rule is:

- add `180.0` degrees to the source longitude
- normalize the result into `[0.0, 360.0)`

This rule is authoritative for:

- Earth from Sun
- South Node from North Node where opposition derivation is used

No alternate opposition rule is allowed.

4.4 Angular Difference Rule

Where this canon requires comparison between two longitudes, angular difference must be treated using canonical circular angle logic rather than naïve subtraction where relevant.

For the Design time solar target search, the authoritative target is based on canonical normalized longitude difference, not on linear raw subtraction outside the circular domain.

4.5 Circular Boundary Rule

Any calculation step involving cyclic angular intervals must use canonical circular wrap handling consistent with:

- normalized longitudes
- half open interval logic where specified
- deterministic boundary ownership
- no double assignment at endpoints

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5. Astrology Calculation Canon

5.1 Astrology Foundation

Astrology in Trinity Engine v1 is calculated using:

- tropical zodiac
- Swiss Ephemeris
- birth moment only
- Placidus house system
- Mean Node as authoritative nodal output policy
- Ascendant and Midheaven as supported angles

No alternate astrology basis is allowed in v1.

5.2 Astrology Raw Object Set

The following raw astrology objects are in scope:

- Sun
- Moon
- Mercury
- Venus
- Mars
- Jupiter
- Saturn
- Uranus
- Neptune
- Pluto
- Chiron
- North Node, Mean
- Earth, derived from Sun plus 180 degrees modulo 360

True Node may also be calculated internally where required for Human Design and Gene Keys support, but it is not part of the authoritative Astrology output object set.

5.3 Astrology Longitude Retrieval

For each supported raw astrology object except Earth, the engine must retrieve or derive the ecliptic longitude at the birth moment according to the pinned astronomical source.

For Earth:

- Earth longitude must be derived as Sun longitude plus 180 degrees modulo 360

Earth must not be independently queried from an alternate astronomical source.

5.4 Astrology Sign Placement Derivation

For each supported astrology object, sign placement must be derived from canonical longitude according to the tropical zodiac division.

Each longitude must map to exactly one sign according to the zodiac interval boundaries defined in the Mapping Canon.

No alternate sign basis is allowed.

5.5 Astrology House Calculation

The engine must calculate house cusps using:

- birth moment
- latitude
- longitude
- Placidus house system

The engine must return:

- house cusps 1 through 12
- Ascendant
- Midheaven

If a supported astrology object is eligible for house placement under the canon rules, the engine must derive the house placement using the canonical house cusp structure.

5.6 Astrology House Placement Rule

For each supported astrology object where house placement applies, the engine must assign exactly one house placement according to the Placidus house structure and the house interval rules defined elsewhere in the canon.

House placement must be based on the canonical longitude of the object and the calculated house cusps.

No alternate house assignment logic is allowed.

5.7 Astrology Node Rule

Astrology uses Mean Node as the authoritative nodal variant.

If both Mean Node and True Node are internally available:

- Mean Node must be used for Astrology output
- True Node must not replace Mean Node in Astrology output
- any use of True Node belongs only to other supported domains where canon explicitly allows it

5.8 Astrology Outputs Explicitly Not Calculated in v1

The following are out of scope for Astrology calculation in v1:

- aspects
- dignities
- dispositors

- house rulers
- elemental balance
- modality balance
- asteroids beyond Chiron
- fixed stars
- any interpretive synthesis

The engine must not derive or return these unless added by later canon revision.

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6. Human Design Snapshot Calculation Canon

6.1 Human Design Foundation

Human Design in Trinity Engine v1 is calculated using:

- the same normalized birth input
- tropical astronomical basis
- True Node as authoritative nodal variant
- a Personality snapshot at birth moment
- a Design snapshot at canon defined design time
- classical Human Design mandala mapping
- canon defined gate and line mapping rules

6.2 Human Design Snapshot Rule

Human Design calculations must always be based on two distinct astronomical snapshots:

1. Personality snapshot
    - calculated at the birth moment
2. Design snapshot
    - calculated at the canon defined design time prior to birth

These two datasets must remain separate throughout calculation and output construction.

They must not be merged into one undifferentiated activation pool.

6.3 Personality Snapshot Object Order

For the Personality snapshot, the canonical object order is:

1. Sun
2. Earth
3. North Node
4. South Node
5. Moon
6. Mercury

7. Venus
8. Mars
9. Jupiter
10. Saturn
11. Uranus
12. Neptune
13. Pluto

This order is authoritative for output, comparison, validation, and downstream derivation.

6.4 Design Time Derivation Mechanics

6.4.1 Purpose

The Design snapshot for Human Design and Gene Keys must be calculated at a second astronomical moment prior to birth called the Design time.

This Design time is derived from solar arc, not from a fixed number of days.

6.4.2 Design Time Target Rule

The Design time target is defined as the moment prior to birth at which the Sun longitude is exactly `88.0` degrees earlier than the birth Sun longitude, using canonical longitude normalization.

Procedure:

1. Determine the Sun longitude at the birth moment.
2. Subtract exactly `88.0` degrees.
3. Normalize the result into `[0.0, 360.0)`.
4. Treat that normalized longitude as the target Sun longitude for the Design time search.

This target is authoritative.

6.4.3 Search Direction Rule

The Design time search must proceed backward in time from the birth moment.

The engine must search only for a valid earlier moment prior to birth.

It must not:

- search forward
- choose the nearest match on either side
- choose a heuristic about 88 days before birth estimate without solving the longitude target

6.4.4 Initial Search Bracket Rule

The initial search bracket must be centered around approximately `88` days before the birth moment.

The bracket must:

- start before the expected solution
- preserve a deterministic bracketing strategy
- be large enough to guarantee deterministic convergence for the supported range

6.4.5 Search Method Rule

The authoritative root finding method for Design time is:

- iterative bisection

No alternate method may be used as the canonical implementation method unless explicitly approved by canon revision.

6.4.6 Stop Conditions

The Design time search must stop when either of the following is true:

1. the absolute Sun longitude difference from the target is less than `0.0001` degrees
2. the remaining time interval width is less than `1` second

6.4.7 Design Time Output Rule

The Design time returned by the search must be treated as the authoritative Design snapshot time in UTC.

All Design snapshot astronomical queries for Human Design and Gene Keys must use this derived Design time.

No later recalculation or alternate adjustment is allowed.

6.4.8 Forbidden Design Time Shortcuts

The engine must not replace the canonical Design time derivation with:

- a fixed `88` day subtraction
- a fixed `3 month` subtraction
- a rough platform imitation
- an average ephemeris approximation
- a precomputed lookup shortcut
- any other heuristic that bypasses the target search

6.5 Design Snapshot Object Order

For the Design snapshot, the canonical object order is the same as the Personality snapshot:

1. Sun
2. Earth
3. North Node
4. South Node
5. Moon
6. Mercury
7. Venus
8. Mars
9. Jupiter
10. Saturn
11. Uranus
12. Neptune
13. Pluto

6.6 Human Design Longitude Policy

For Human Design calculations:

- True Node is authoritative
- South Node must be derived as the canonical opposite of North Node where applicable
- Earth must be derived as the canonical opposite of Sun where applicable

The astronomical layer used for Human Design must be consistent across both Personality and Design snapshots.

6.7 Personality and Design Activation Outputs

For each snapshot object, the engine must return:

- object identifier
- gate
- line
- ordered activation record

The engine must preserve snapshot identity:

- Personality activations remain Personality activations
- Design activations remain Design activations

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7. Human Design and Gene Keys Gate and Line Derivation Mechanics

7.1 Mandala Foundation

Gate and line derivation for Human Design and Gene Keys must use the canonical Human Design mandala.

The authoritative mandala parameters are:

- canonical gate sequence anchor: as defined in Document 06
- gate width: `5.625°`
- line width: `0.9375°`
- direction: ascending ecliptic longitude
- interval rule: start inclusive, end exclusive
- gate order: fixed lookup list, not calculated procedurally

These values are authoritative.

Any superseded or previously rejected anchor assumption must not be used. The authoritative canonical mandala anchor is defined in Document 06.

7.2 Gate Mapping Input

For each Human Design object in each snapshot:

- obtain the canonical normalized longitude
- use that longitude as the sole input to gate mapping

The same rule applies to Gene Keys because Gene Keys derives directly from Human Design gate and line outputs.

7.3 Relative Mandala Position Rule

To map a longitude to a gate:

1. normalize the longitude into `[0.0, 360.0)`
2. subtract the canonical mandala anchor defined in Document 06
3. normalize the result into `[0.0, 360.0)`

The resulting value is the relative mandala position.

This relative mandala position is the authoritative angular input for gate and line assignment.

The first gate segment in the canonical sequence begins at the pinned angular anchor defined in Document 06, and all downstream gate and line assignment must use that anchor consistently.

7.4 Gate Index Rule

Gate assignment must be derived from the relative mandala position by:

1. dividing the relative mandala position by the canonical gate width `5.625°`
2. taking the mathematical floor of the result
3. using that integer as the zero based gate segment index
4. resolving that segment index through the pinned fixed gate order lookup list

The gate order list itself must be defined in Document 06.

No implicit numeric gate sequence may be assumed.

7.5 Line Index Rule

Line assignment must be derived from the relative position within the assigned gate interval by:

1. determining the relative offset inside the gate segment
2. dividing that offset by the canonical line width `0.9375°`
3. taking the mathematical floor of the result
4. converting that zero based value into the canonical line number according to the pinned line numbering scheme

For v1, the line representation is `1` through `6`.

7.6 Interval Inclusion Rule

Gate and line intervals are half open intervals:

- interval start is inclusive
- interval end is exclusive

This means:

- a value exactly on an interval start belongs to that interval
- a value exactly on an interval end belongs to the next interval

This rule is authoritative for:

- gate assignment
- line assignment
- boundary handling
- wraparound behavior at `0.0°`

7.7 Boundary Rule

Boundary cases must always resolve into the next segment when the canonical interval end is hit.

This rule prevents:

- ambiguous segment ownership

- double assignment
- platform dependent boundary drift

7.8 Gate and Line Determinism Rule

Gate and line assignment must use:

- canonical normalized longitude
- canonical mandala anchor as defined in Document 06
- canonical fixed gate width
- canonical fixed line width
- canonical fixed gate order list
- canonical half open interval rule
- floor operations

No alternate segmentation logic, rounding shortcut, platform imitation, or superseded anchor assumption is allowed.

7.9 Gene Keys Mapping Identity Rule

Gene Keys must use the exact same gate and line results as Human Design.

The engine must not:

- recalculate Gene Keys gates independently
- apply a separate mandala
- apply a separate line system
- apply any semantic mutation to the gate or line identifiers

For v1:

- Human Design gate = Gene Keys key identifier
- Human Design line = Gene Keys line

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8. Human Design Structural Derivation Mechanics

8.1 Purpose of This Section

This section defines the deterministic derivation rules for Human Design structural outputs in Trinity Engine v1.

Its purpose is to specify the canonical rules for deriving:

- Channels
- Centers

- Definition
- Type
- Authority
- Incarnation Cross encoding

These derivations must be computed from the canon defined Human Design activation set and must not rely on interpretation, platform imitation, or informal community shorthand.

8.2 Combined Activation Set Rule

Unless a rule explicitly states otherwise, Human Design structural derivations must use the combined Human Design activation set, defined as:

- all canon valid Personality activations
- all canon valid Design activations

An activation is considered available to the combined set if its gate has been validly derived from the canon defined gate and line mapping process.

Line values do not affect channel activation, center definition, definition class, Type, or Authority unless explicitly stated elsewhere in the canon.

8.3 Channel Derivation Rule

A Human Design channel is defined only when both gates of a canon valid channel pair are active in the combined Human Design activation set.

A channel must not be considered active when:

- only one gate of the pair is active
- the gate pair is not present in the canonical channel table in Document 06
- the gates are active in separate systems outside the combined Human Design activation set

For each active channel, the engine must return a stable channel identifier derived from the canonical channel table.

A channel must be returned at most once, even if its two gates are activated multiple times across Personality and Design.

Duplicate activation sources do not create duplicate channels.

8.4 Center Derivation Rule

Human Design centers are derived from active channels.

A center is defined if it participates in at least one active canon valid channel.

A center is undefined if it participates in no active canon valid channel.

The authoritative center list and channel to center mapping are defined in Document 06.

For each canon supported center, the engine must return:

- center identifier
- state: defined or undefined

No gradient, percentage, or partial definition model is allowed in v1.

8.5 Definition Derivation Rule

Definition must be derived from the connectivity graph formed by:

- all defined centers
- all active channels connecting those centers

Each group of defined centers connected through one or more active channels constitutes one connected component.

The total number of connected components in the defined center graph determines the Definition class.

Trinity Engine v1 supports the following Definition classes only:

- `none`
- `single`
- `split`
- `triple_split`
- `quadruple_split`

Assign Definition as follows:

- `none` if there are no defined centers
- `single` if all defined centers belong to one connected component
- `split` if defined centers form exactly two disconnected connected components
- `triple_split` if defined centers form exactly three disconnected connected components
- `quadruple_split` if defined centers form exactly four disconnected connected components

No other Definition class is supported in v1 unless explicitly added by later canon revision.

8.6 Type Derivation Rule

Type must be derived from:

- defined center states
- active channel set
- motor to throat connectivity
- Definition state where relevant

For Trinity Engine v1, the authoritative motor centers are:

- `root`
- `sacral`
- `solar_plexus`
- `ego`

A motor is considered connected to the Throat if:

- there is at least one active channel directly connecting that motor center to the Throat
- or there is a continuous path of active channels linking that motor center to the Throat through defined centers

Type must be assigned in the following order:

Rule 1. Reflector

If Definition is `none`, assign:

- `reflector`

Rule 2. Generator Family

If the `sacral` center is defined, assign within the Generator family as follows:

- if at least one motor to Throat connectivity path exists, assign:
  - `manifesting_generator`
- otherwise assign:
  - `generator`

Rule 3. Manifestor

If the `sacral` center is undefined and at least one non `sacral` motor center is connected to the Throat, assign:

- `manifestor`

Rule 4. Projector

If none of the above rules apply and Definition is not `none`, assign:

- `projector`

These assignment rules are exhaustive for Trinity Engine v1.

8.7 Authority Derivation Rule

Authority must be assigned deterministically from the structural Human Design state.

Authority must not be inferred from narrative meaning, interpretation, or external platform labels.

Authority must be assigned in the following priority order:

Rule 1. Emotional Authority

If the `solar_plexus` center is defined, assign:

- `emotional`

Rule 2. Sacral Authority

If Emotional Authority is not assigned and the Type is `generator` or `manifesting_generator`, assign:

- `sacral`

Rule 3. Splenic Authority

If no higher priority authority has been assigned and the `spleen` center is defined, assign:

- `splenic`

Rule 4. Ego Manifested Authority

If no higher priority authority has been assigned, the Type is `manifestor`, and the `ego` center is defined, assign:

- `ego_manifested`

Rule 5. Ego Projected Authority

If no higher priority authority has been assigned, the Type is `projector`, and the `ego` center is defined, assign:

- `ego_projected`

Rule 6. Self Projected Authority

If no higher priority authority has been assigned, the Type is `projector`, and the `g` center is defined, assign:

- `self_projected`

Rule 7. Mental Authority

If no higher priority authority has been assigned and the Type is `projector`, assign:

- `mental`

Rule 8. Lunar Authority

If the Type is `reflector`, assign:

- `lunar`

The first matching rule in the assignment order is authoritative.

8.8 Profile Derivation Rule

Profile must be derived from:

- Personality Sun line
- Design Sun line

The canonical Profile output format must be:

- `personality_line/design_line`

Example:

- `3/5`

No alternate ordering is allowed.

8.9 Incarnation Cross Encoding Rule

Incarnation Cross must be derived from:

- Personality Sun
- Personality Earth
- Design Sun
- Design Earth

For Trinity Engine v1, the required canonical representation of Incarnation Cross is a deterministic structural encoding composed of these four activations.

At minimum, the engine must return:

- Personality Sun gate and line
- Personality Earth gate and line
- Design Sun gate and line
- Design Earth gate and line

A human readable Incarnation Cross name is not required at the calculation layer unless a later canon document pins an authoritative naming table.

8.10 Out of Scope Structural Behaviors

The following are out of scope for Trinity Engine v1 structural derivation:

- strategy text
- not self themes
- signature text
- variable derivation
- arrows
- digestion
- cognition
- environment
- motivation
- view
- perspective
- any interpretive BodyGraph commentary

These belong outside the v1 calculation layer.

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9. Gene Keys Calculation Canon

9.1 Gene Keys Foundation

Gene Keys in Trinity Engine v1 is not an independent astronomical system.

It must be derived strictly from Human Design gate and line outputs.

No separate astronomical query, separate nodal policy, or alternate mapping basis is allowed.

9.2 Supported Gene Keys Outputs

Gene Keys in v1 supports:

- key identifier
- line

- Activation Sequence positions only:
    - Life's Work
    - Evolution
    - Radiance
    - Purpose

9.3 Gene Keys Identifier Rule

For each supported Gene Keys position:

- the key identifier must match the corresponding Human Design gate identifier
- the line must match the corresponding Human Design line identifier

No alternate numbering or alternate line derivation is allowed.

9.4 Activation Sequence Derivation Rule

The Activation Sequence positions must be derived as follows:

- Life's Work = Personality Sun
- Evolution = Personality Earth
- Radiance = Design Sun
- Purpose = Design Earth

No additional Gene Keys sequence positions are in scope for v1.

9.5 Gene Keys Content Boundary

The engine must return only identifiers for Gene Keys outputs.

The engine must not generate:

- shadow text
- gift text
- essence text
- sequence prose
- semantic states
- interpretive or narrative content

These belong exclusively to the content layer outside Trinity Engine v1.

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10. Internal Combination Rules

10.1 Combined Human Design Activation Set

Where a Human Design structural derivation requires the combined activation set, the combined set must consist of:

- all valid Personality activations
- all valid Design activations

No activation may be dropped unless explicitly excluded by canon rule.

10.2 Oppositional Object Derivation Rule

Where Earth or South Node are derived by opposition:

- the derivation must use canonical 180 degree opposition logic
- the result must remain normalized within canonical longitude bounds

No alternate opposition method is allowed.

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11. Rounding and Boundary Rules

11.1 General Rule

Any rounding, normalization, interval inclusion rule, endpoint rule, or tie breaking behavior that can affect:

- sign assignment
- house assignment
- gate assignment
- line assignment
- design time determination
- structural derivations

must be explicitly pinned in the final canon.

11.2 Current Pinned Boundary Rules

The following boundary behaviors are already pinned:

- longitude domain is `[0.0, 360.0)`
- `360.0` normalizes to `0.0`
- gate and line intervals are start inclusive and end exclusive
- interval end belongs to the next segment
- Design time search uses bisection
- Design time stop conditions are:
    - absolute Sun longitude difference less than `0.0001` degrees
    - or remaining interval width less than `1` second

11.3 Remaining Dependency Rule

Any additional boundary behavior required by channel, center, definition, Type, Authority, or output encoding must be made explicit in later documents if not already pinned here.

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12. Out of Scope Calculation Behavior

The engine must not calculate or derive:

- astrology aspects
- advanced astrology interpretive constructs
- Human Design Variables
- arrows
- digestion
- cognition
- environment
- motivation
- view
- perspective
- Gene Keys Venus Sequence
- Gene Keys Pearl Sequence
- Gene Keys semantic text states
- any content layer output

unless explicitly added by later canon revision.